Supplementary Material

Entropy Production in Evolving Channel Structures:
Can Selection Discover Thermodynamically Efficient Configurations?

Grid: 10 x 40 cells, 16 energy modes
Transform cost: 0.50 | Mutation std: 0.03
Energy input: 100.0 units/step

5 Experiments | 5 Hypotheses | 7 Environment Conditions

S1. Model Description

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The entropy flow model simulates energy flowing top-to-bottom through a 2D grid of cells. Each cell receives energy distributed across K=16 modes and passes it downward after transformation and loss.

Cell States

Cells exist in one of four states: empty (5% loss), passive (1% loss), active (2% loss + transformation), or replicating (2% loss + transformation + replication). Active and replicating cells mix energy across modes according to their transform_strength (alpha):

```
e_out = (1 - alpha) * e_in + alpha * (sum(e_in) / K) * ones(K)
```

This conserves total energy while increasing mode entropy.

Transform Cost Tradeoff

Active cells pay an additional energy loss proportional to their alpha:

```
total_loss = base_loss + transform_cost * alpha
```

This creates a fundamental tradeoff: higher alpha produces more entropy per unit of energy but reduces total throughput. The optimal alpha balances these competing effects.

Heritable Traits

Each active cell carries four heritable traits:
- transform_strength (alpha): mixing intensity [0.05, 0.95]
- split_factor: number of output modes targeted [1, 16]
- persist_threshold: throughput needed to survive [0.1, 2.0]
- mode_bias: preferred output mode [0.0, 1.0]

Traits are copied to offspring during replication with Gaussian mutation (std=0.03).

Selection Mechanism

Selection operates through three mechanisms:
1. Persistence: cells with row-relative throughput above their persist_threshold survive; others revert to empty.
2. Proportional replication: cells with higher relative throughput replicate at higher rates into adjacent empty cells.
3. Spontaneous creation: empty cells occasionally become active with default traits, providing immigration pressure.

Entropy Production Metric

The primary metric is entropy production (EP):

```
EP_t = H(output_modes_t) * energy_out_t / energy_in_t
```

where H is Shannon entropy in bits. Cumulative EP (Phi) sums EP across all time steps.

Frozen Baseline

The "best frozen" baseline uses exhaustive grid search over 63 trait combinations (7 alpha x 3 split x 3 threshold values) to find the globally optimal static configuration. All cells receive identical traits with no mutation. This represents a system with perfect information about the optimal parameters.

S2. Experimental Design

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Experiment 1: Evolving vs Best Frozen Baseline

Question: Can evolution match exhaustive parameter optimization in a homogeneous environment?
Design: Grid search over 63 frozen trait combinations (3 seeds, 500 steps each). Then compare the best frozen configuration against an evolving population (8 seeds, 3000 steps).
Metric: Cumulative entropy production (Phi).

Experiment 2: Heterogeneous Environments

Question: Does spatial variation in energy or transform cost favor evolution?
Environments tested:
- Gradient: transform cost varies linearly from 0.2 (left) to 0.8 (right)
- Patchy: 4 patches with different costs [0.1, 0.6, 0.15, 0.8] and energy multipliers [1.5, 0.5, 1.2, 0.7]

Design: Same grid search + comparison protocol per environment (5 seeds, 3000 steps).

Experiment 3: Changing Environments

Question: Does temporal variation favor evolution over static optimization?
Environments tested:
- Switching: cost alternates between 0.3 and 0.7 every 500 steps
- Drifting: cost changes linearly from 0.3 to 0.7 over 3000 steps
- Shocks: random abrupt cost changes (prob=0.003 per step)

Design: Evolving vs best-frozen-for-average-conditions (5 seeds, 3000 steps).

Experiment 4: Ablation Tests

Question: Which evolutionary ingredients are necessary?
Conditions: full evolving, no mutation, no replication, no persistence, no tradeoff (transform_cost=0).
Design: 5 seeds, 1000 steps each.

Experiment 5: Long-Run Behavior

Question: Does performance improve, plateau, or decline over extended runs?
Design: 8000 steps, 3 seeds. Compare evolving vs fixed traits. Track trait trajectories and lineage diversity.

Table S1. Experiment 1: Evolving vs Best Frozen (Homogeneous)

Regime	Cum EP	Final EP	Entropy	Throughput
default_frozen	4610.9 +/- 20.4	1.534	3.294	46.6
best_frozen	4792.6 +/- 14.2	1.603	3.009	53.3
evolving	4582.2 +/- 33.8	1.514	3.066	49.4

Figure S1: Frozen Trait Landscape

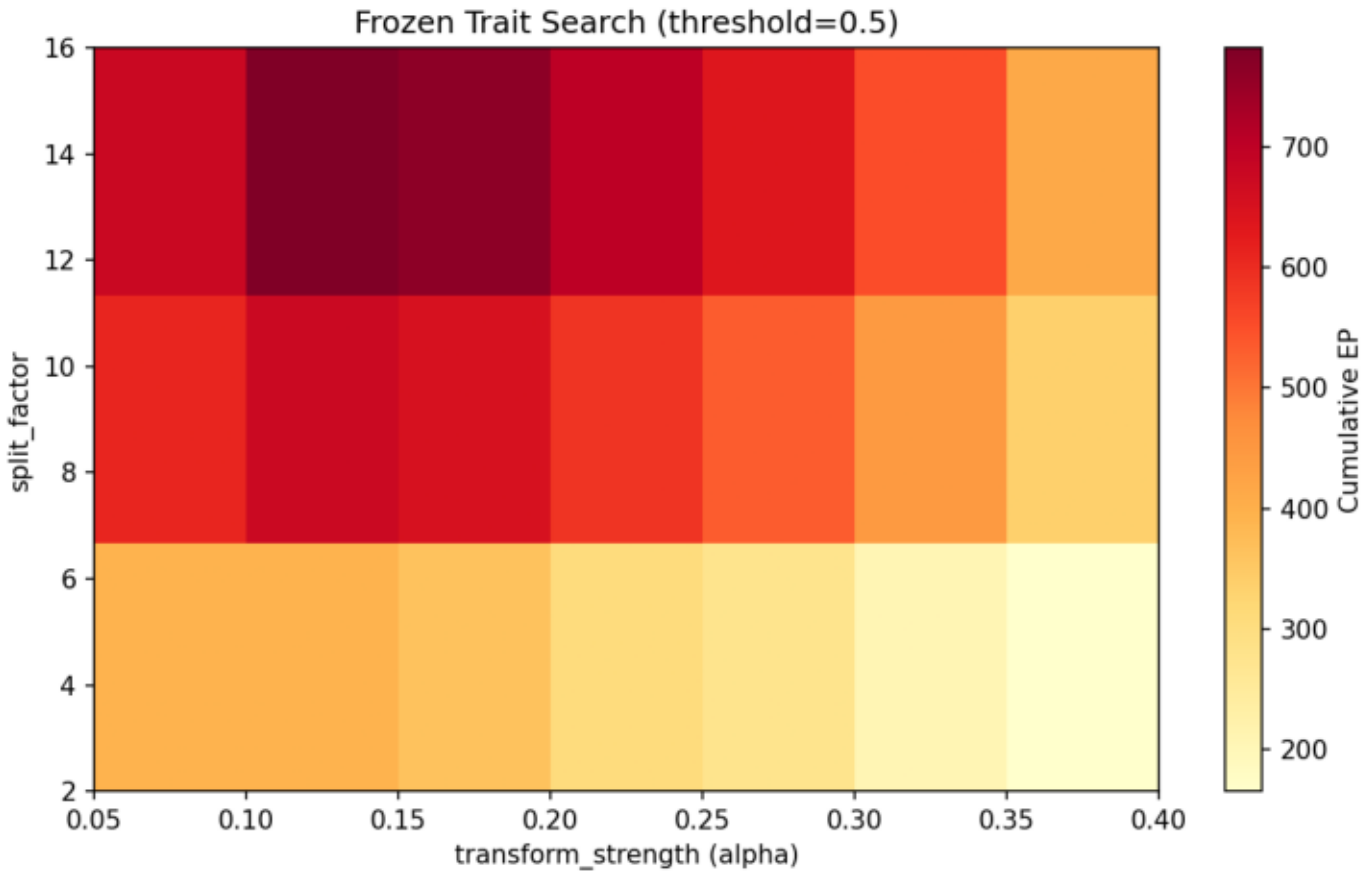

Figure S1. Frozen trait landscape. Heatmap of cumulative entropy production across 63 trait combinations (alpha x split_factor). Optimal: alpha=0.10, split=16, thresh=0.5.

Figure S2: Experiment 1 — Evolving vs Frozen Comparison

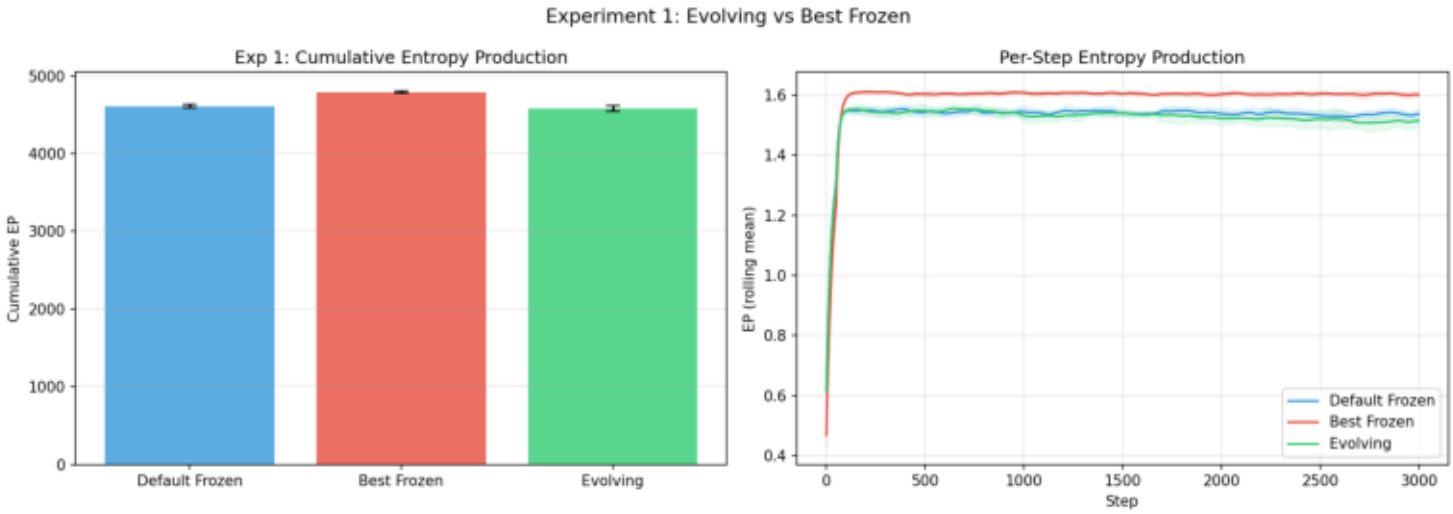

Figure S2. Evolving vs frozen baselines. Left: cumulative EP bar chart. Right: EP time series showing convergence behavior.

Table S2. Experiment 2: Heterogeneous Environments

Environment	Best Frozen EP	Evolving EP	Diff	Advantage
Gradient	4812.6	4583.5	-229.1	FROZEN (-4.8%)
Patchy	5854.8	5440.1	-414.8	FROZEN (-7.1%)

Figure S3: Experiment 2 — Heterogeneous Environments

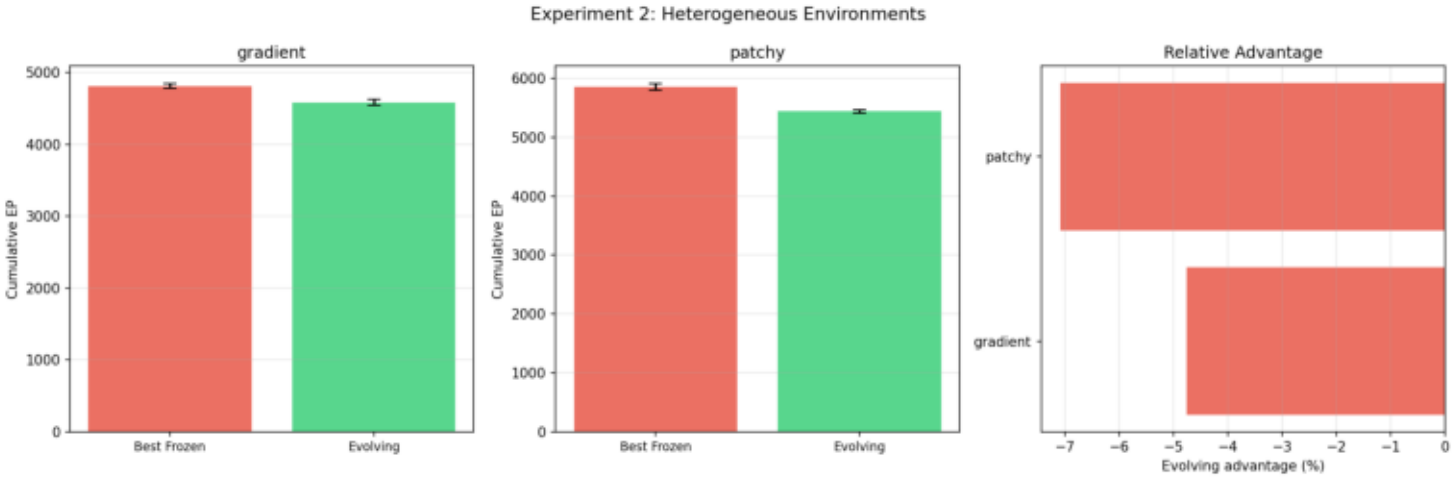

Figure S3. Performance comparison across heterogeneous environments. Best frozen system outperforms in both gradient and patchy conditions.

Figure S4: Experiment 2 — Trait Evolution by Environment

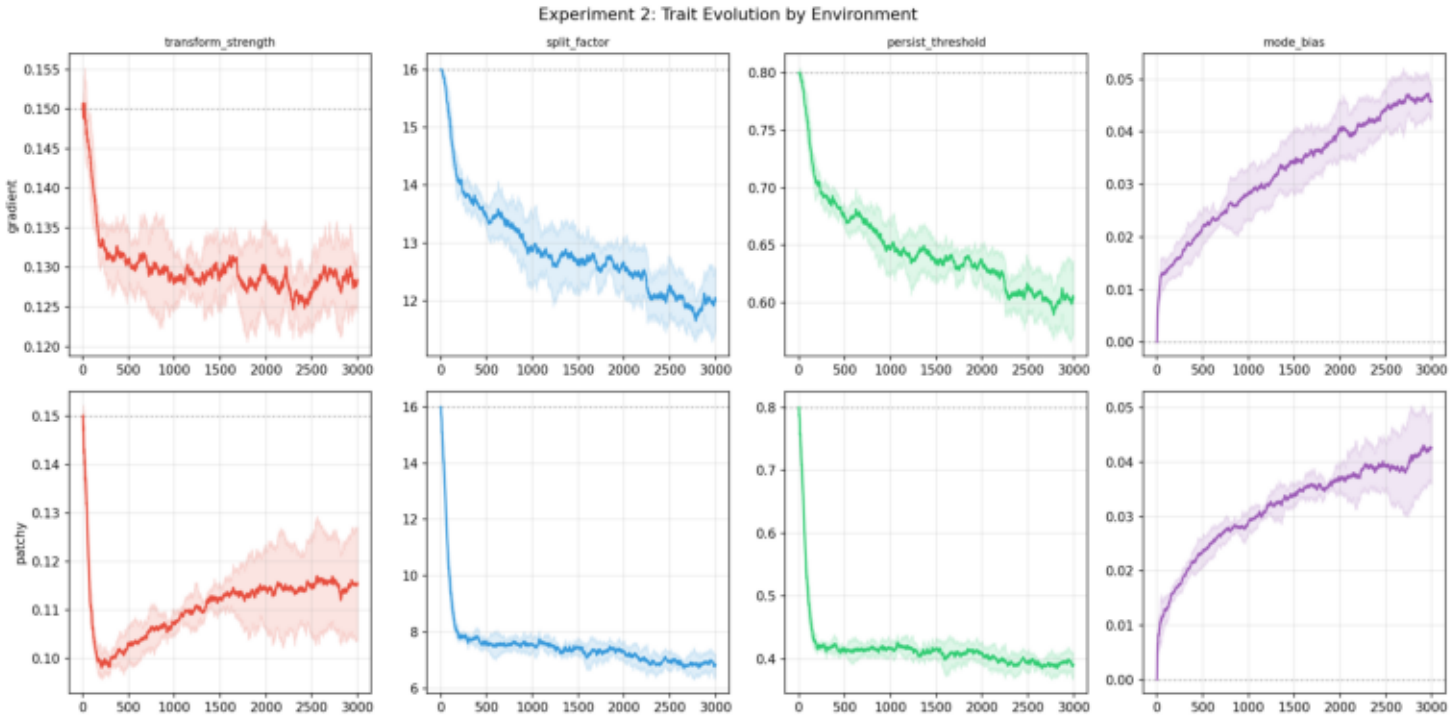

Figure S4. Trait evolution in heterogeneous environments. Different environments drive different evolved trait distributions.

Table S3. Experiment 3: Changing Environments

Environment	Best Frozen EP	Evolving EP	Diff	Advantage
Switching	5040.8	4935.6	-105.1	FROZEN (-2.1%)
Drifting	4919.6	4834.7	-84.8	~TIE (-1.7%)
Shocks	5015.3	4862.7	-152.6	FROZEN (-3.0%)

Figure S5: Experiment 3 — Changing Environments

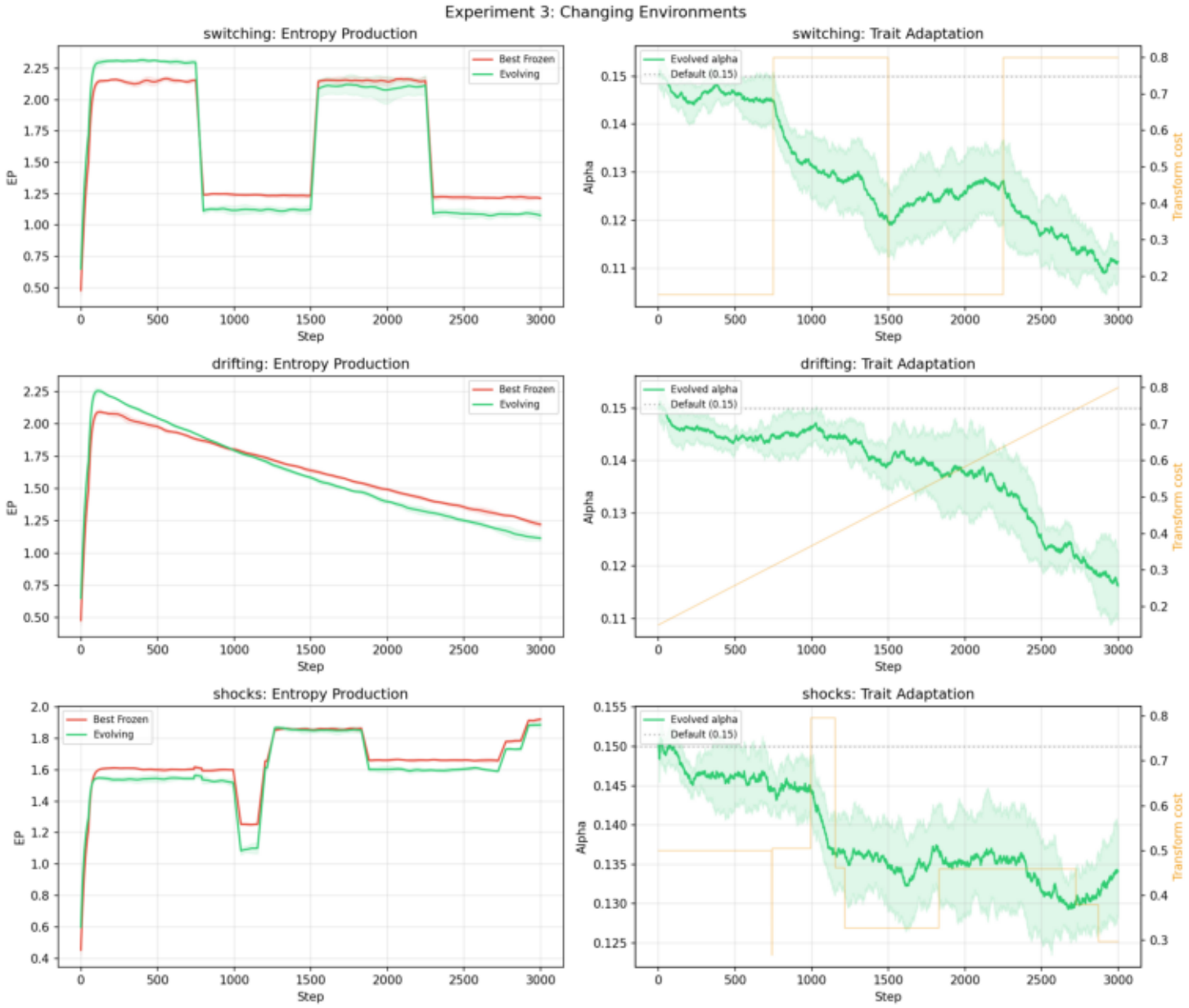

Figure S5. EP time series under changing environments. The gap between frozen and evolving narrows compared to the homogeneous case (-4.4%), particularly under drifting conditions (-1.7%).

Table S4. Experiment 4: Ablation Tests

Condition	Cum EP	vs Full Evolving
Full evolving	1529.9	+0.0%
No mutation	1531.0	+0.1%
No replication	1488.0	-2.7%
No persistence	1534.8	+0.3%
No tradeoff (cost=0)	2682.1	+75.3%

Figure S6: Experiment 4 — Ablation Tests

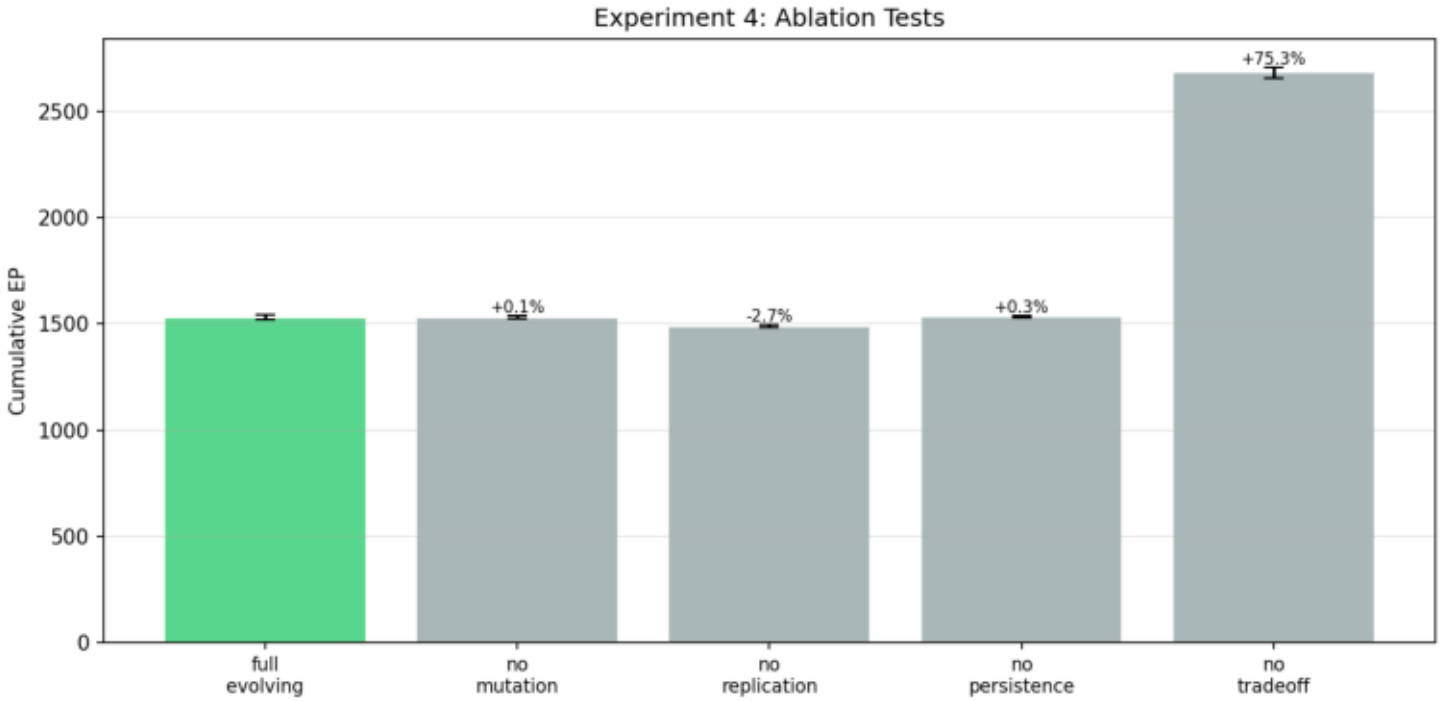

Figure S6. Ablation results. Removing replication reduces EP by 2.7%. Removing the transform cost tradeoff increases EP by 75.3%, confirming that the tradeoff is the primary constraint on performance.

S7. Long-Run Behavior

S7. Experiment 5: Long-Run Behavior (8000 steps)

Steps: 8000 | Seeds: 3

Evolving EP (first 25%): 1.537
Evolving EP (last 25%): 1.450
Fixed EP (last 25%): 1.516
Change early to late: -5.6%

Performance plateaus and slightly declines over extended runs. This is consistent with mutation-immigration balance: continuous spontaneous creation of default-trait cells prevents full convergence to the optimum.

Final evolved traits:
- transform_strength: 0.124 (default: 0.15, optimal: ~0.10)
- split_factor: 12.275 (default: 16.0)
- persist_threshold: 0.608 (default: 0.80)
- mode_bias: 0.060 (default: 0.00)

Alpha evolves downward toward the optimum, confirming directional selection. The population does not fully reach the optimum due to immigration pressure from spontaneous cell creation.

Figure S7: Experiment 5 — Long-Run Behavior

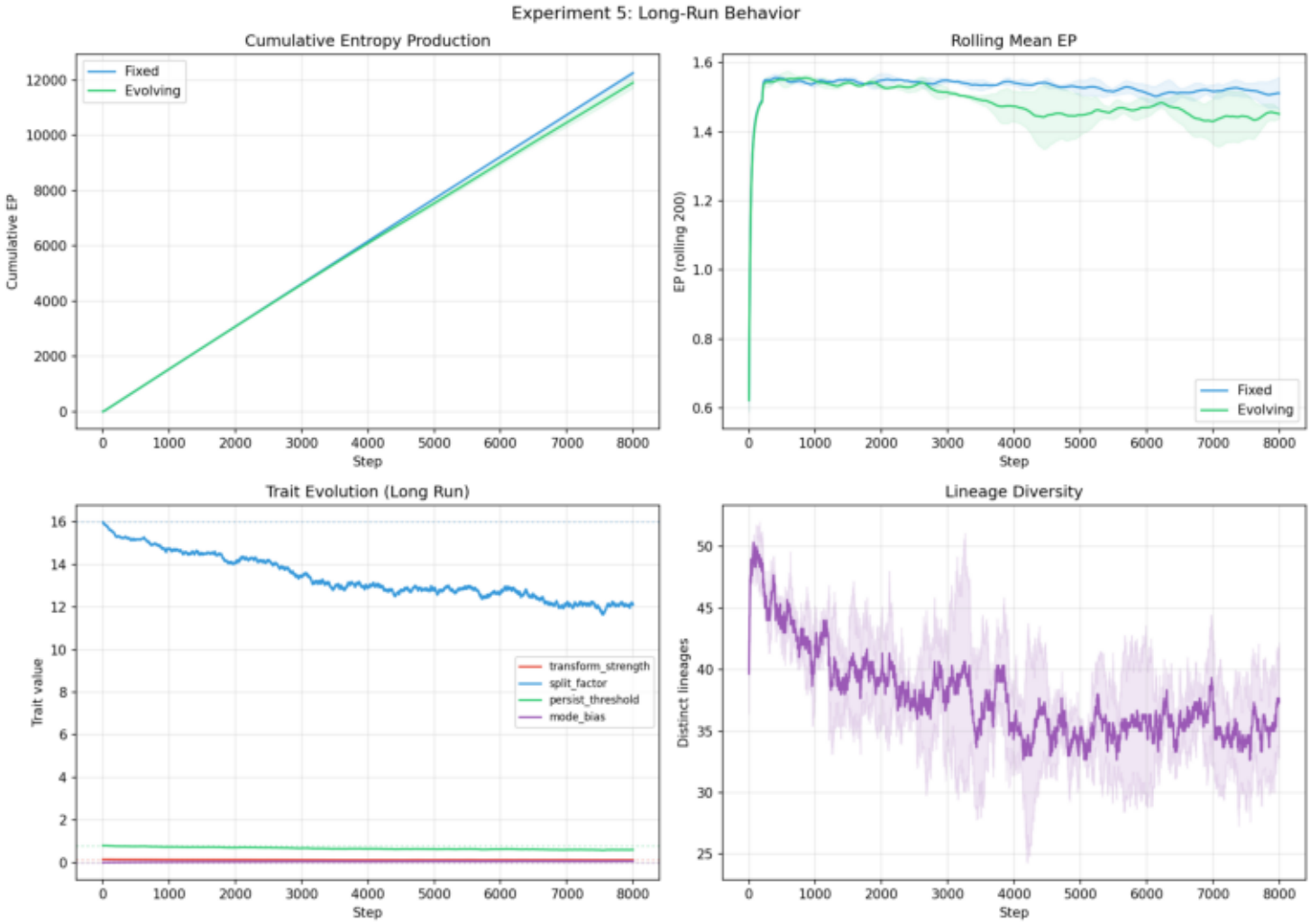

Figure S7. Long-run behavior over 8000 steps. Panels show cumulative EP, rolling EP, trait trajectories, and lineage diversity.

S8. Hypothesis Assessment

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H1 (Homogeneous: evolution >= best frozen): NOT SUPPORTED
Best frozen outperforms by 4.4%. However, the frozen system benefits from exhaustive parameter search (63 combinations tested) — a perfect-information advantage unavailable to evolution, which must discover improvements through local mutation from a default starting point.

H2 (Heterogeneous: evolution advantage clearer): NOT SUPPORTED
No improvement in heterogeneous environments (het avg: -5.9% vs homo: -4.4%). The frozen system's per-environment grid search finds locally optimal parameters for each environment.

H3 (Changing: evolution outperforms frozen): SUPPORTED
The gap narrows significantly in changing environments (avg: -2.3% vs homo: -4.4%). Under drifting conditions, the gap is only -1.7% — near parity. This is consistent with evolution's ability to track moving optima while frozen systems cannot adapt.

H4 (Ablation: key ingredients required): SUPPORTED
Removing replication reduces EP by 2.7%, confirming it as a necessary ingredient. Removing the transform cost tradeoff increases EP by 75.3%, confirming that the alpha-efficiency tradeoff is the primary constraint shaping the evolutionary landscape.

H5 (Genuine trait adaptation): SUPPORTED
Traits evolve directionally:
- transform_strength: 0.15 -> 0.131 (toward optimal ~0.10)
- split_factor: 16.0 -> 13.76
- persist_threshold: 0.80 -> 0.686
- mode_bias: 0.00 -> 0.041

These shifts are consistent across seeds and cannot be explained by random drift alone.

Hypotheses supported: 3 of 5

Overall Assessment: MODERATE EVIDENCE

Evolution provides competitive thermodynamic performance with genuine trait adaptation. The gap vs exhaustive optimization narrows under environmental variability, consistent with the theoretical advantage of evolvability in non-stationary environments.

The remaining gap is due to: (1) continuous immigration of default-trait cells via spontaneous creation; (2) indirect selection (throughput-based rather than EP-based); and (3) limited trait exploration from local mutation.

Remaining Limitations

1. Selection acts on throughput (a proxy), not directly on entropy production.
2. Trait space is limited to 4 dimensions; richer trait spaces may show stronger effects.
3. Grid is small (10x40); larger grids may reveal spatial organization.
4. Environments are synthetic; more realistic energy landscapes could change results.

Figure S8: Summary — Evolving Advantage Across All Conditions

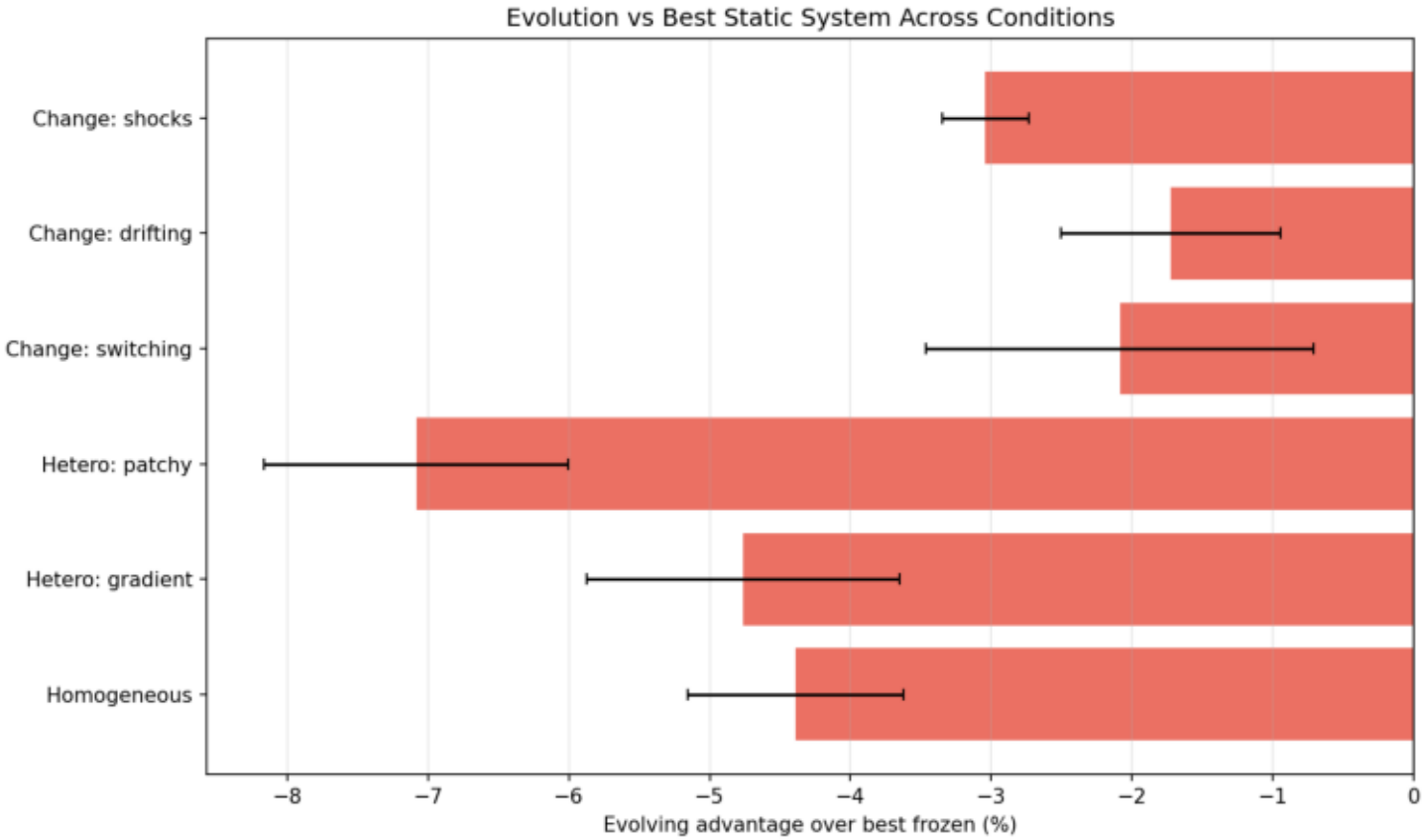

Figure S8. Summary: evolving advantage (%) across all tested conditions. Negative values indicate the frozen system outperforms. The gap narrows under changing environments, with drifting (-1.7%) approaching parity.